

AI CAREER FOR WOMEN

Engineering Spoke



FemCare- Smart Pregnancy Care Companion



A Project Report
submitted in partial fulfillment of the requirements
of
AICW project.

By

Gandham Sirisha, gandhamsirisha709@gmail.com

Karimireddy Aswini Sowmya,

ashwinikarimireddy09@gmail.com

Ubalanka Kalavathi, kalavathiubalanka@gmail.com

Gandhikota Jahnavi, januists23@gmail.com

Under the Guidance of
M Bharathi Raja

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ABSTRACT

This project focuses on the development of **Pregbot – Smart Pregnancy Care Companion**, which aims to address the challenges faced by expectant mothers in accessing reliable pregnancy-related information, overcoming language barriers, and receiving continuous health guidance. The primary objective of this system is to provide personalized pregnancy care, reminders, and awareness through an intelligent chatbot-based platform.

To achieve this, the system integrates **Machine Learning (ML)** and **Natural Language Processing (NLP)** techniques along with modern web technologies. The application is developed using **Python with Flask** as the backend framework, **MySQL** as the database, and **HTML, CSS, and JavaScript** for the frontend. A **feed-forward neural network** is implemented to classify user queries into predefined pregnancy-related categories, enabling accurate and relevant responses. Additionally, **multilingual support** is provided using the Google Translate API, allowing users to interact in multiple regional languages.

The system delivers various pregnancy-related services such as diet recommendations, exercise guidance, medication reminders, and general healthcare information through an interactive chatbot interface. The system is tested using sample user inputs in a controlled environment to evaluate its performance and response accuracy.

The results demonstrate that the proposed system improves accessibility, efficiency, and user interaction compared to traditional methods of information delivery. This project can be applied in maternal healthcare systems, awareness platforms, and rural healthcare support. Furthermore, future enhancements may include voice-based interaction and integration with real-time health monitoring devices.

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CHAPTER 1

INTRODUCTION

Women's health and safety are important aspects of society that require continuous attention and support. In today's digital world, technology plays a vital role in providing solutions for health monitoring and personal safety. However, many women still face challenges such as lack of awareness about health issues, difficulty in tracking health conditions, and limited access to emergency support systems.

Traditional methods of health tracking, such as manual recording, are often inaccurate and difficult to maintain. Similarly, in emergency situations, women may not have quick access to help or communication tools to ensure their safety. Existing applications usually focus on either health tracking or safety features but fail to provide a complete and integrated solution.

To address these challenges, this project proposes **FemCare: Women Health and Safety Support System**. The system is designed to provide a unified platform where women can track their health, receive timely reminders, access useful health information, and send emergency alerts when needed. By combining health monitoring and safety features into a single application, FemCare aims to improve convenience, awareness, and security.

The system uses modern web technologies to ensure ease of use, accessibility, and reliability. It helps users manage their daily health activities while also providing support during critical situations. The integration of multiple features into one platform makes the system more efficient and user-friendly. In conclusion, FemCare is developed as a practical solution to enhance women's health awareness and ensure personal safety through a simple and effective digital platform.

1.1 Problem Statement

Women face several challenges related to health management and personal safety. Many existing systems lack proper integration of health tracking and emergency support. Manual tracking methods are inefficient, and in critical situations, immediate help may not be easily accessible. Therefore, there is a need for a system that provides reliable health monitoring and quick emergency assistance in a single platform.

Motivation

The motivation behind developing the **FemCare: Women Health and Safety Support System** arises from the increasing need to address challenges related to women's health awareness and personal safety in today's society.

Many women face difficulties in maintaining proper health records, especially for menstrual cycles and related health conditions. Manual tracking methods are often inconvenient, error-prone, and easily forgotten. This creates a need for a digital system that can provide accurate tracking and timely reminders.

Another important factor is the growing concern for women's safety. In emergency situations, women may not have immediate access to help or may not be able to contact others quickly. This highlights the need for a system that can send instant alerts and provide quick assistance during critical situations.

Lack of awareness about women's health is also a major issue. Many individuals do not have

access to proper health guidance or reliable information. Providing health tips and educational content through a digital platform can help improve awareness and promote better health practices.

1.1 Objective

The main objectives of this project are:

- To develop a system for **women's health tracking**
- To provide **reminders for health activities**
- To implement an **emergency alert system**
- To give **health tips and awareness**
- To create a **simple and user-friendly application**
- To combine health and safety features in **one platform**

1.2 Scope of the Project

The FemCare system is designed to help women manage their health and ensure their safety in daily life. It allows users to track their health conditions, receive reminders, and access useful health information.

The system also provides an emergency alert feature, which helps users send quick alerts during unsafe situations. It can be used by individuals for personal health monitoring and safety support.

The project is mainly developed as a web-based application and can be used anywhere with an internet connection. In the future, the system can be expanded by developing a mobile application, adding AI-based features, and integrating real-time health monitoring.

CHAPTER 2

LITERATURE SURVEY

2.1 Review of Relevant Literature

In recent years, many applications have been developed to support women's health and safety. Health tracking applications help users monitor menstrual cycles, fitness, and overall well-being. These systems improve awareness and help users maintain better health records. Similarly, safety applications provide features such as emergency alerts, location sharing, and quick communication during unsafe situations. These applications aim to ensure user safety and provide immediate assistance when needed.

Some systems also provide health tips and awareness through digital platforms. These solutions help users gain knowledge about health-related issues and improve their lifestyle.

However, most of these systems focus only on a single feature, either health tracking or safety, and do not provide a combined solution.

2.2 Existing Models, Techniques, and Methodologies

Several technologies are used in existing systems, such as:

- Mobile Applications for health tracking and safety alerts
- Reminder Systems for tracking health activities
- Location-Based Services for emergency support
- Web Technologies like HTML, CSS, JavaScript for user interface
- Database Systems for storing user data

These systems are designed to provide basic functionalities but are often limited in integration and performance.

2.3 Limitations of Existing Solutions and Research Gap

Although many applications are available, they have some limitations:

- Lack of integration between health tracking and safety features
- Limited awareness and guidance features
- Dependence on multiple applications for different needs
- Some systems are not user-friendly

There is a need for a system that combines all essential features into a single platform. The proposed FemCare system addresses this gap by integrating health tracking, emergency alerts, and awareness features in one application.

CHAPTER 3

PROPOSED METHODOLOGY

The proposed system, **FemCare: Women Health and Safety Support System**, is designed to provide a complete solution for women's health tracking and safety. The system integrates features such as health monitoring, reminders, emergency alerts, and awareness into a single platform. The system is developed as a web-based application where users can easily access all features through a simple interface. It ensures convenience, reliability, and quick access to important functionalities.

3.1 System Design

The proposed system follows a structured design that integrates **user registration, health tracking, reminder system, and emergency alert mechanisms**. The system ensures that the user can easily access all features and receive support during both normal and emergency situations..



3.1.1 Registration

The registration module allows users to create an account in the FemCare system by providing basic personal details. This step ensures that only authorized users can access the system features. During registration, the user enters information such as name, email ID, and mobile number. The system stores this data securely in the database and creates a unique user account.

Steps involved:

- User enters personal details (name, email, phone number)
- System validates the entered information
- Data is stored in the database
- User account is successfully created

This module ensures secure access and enables users to use features like health tracking, reminders, and emergency alerts.

3.1.2 Recognition

The recognition module is used to verify the user before allowing access to the FemCare system. It ensures that only registered users can log in and use the application features.

In this system, recognition is done through user login credentials such as email and password. The system checks the entered details with the stored data in the database. If the details match, the user is successfully authenticated and allowed to access the dashboard.

Steps involved:

- User enters login credentials (email and password)
- System verifies the details with the database
- If details match, access is granted
- If details do not match, login is denied

This module helps in maintaining security and prevents unauthorized access to the system.

3.2 Modules Used

The user module is the main part of the system where users can access all features.

Functions:

- Register and login
- Track menstrual cycle and health data
- Set reminders for important dates
- Send emergency alerts
- View health tips and information

Admin Module

The admin module is responsible for managing the overall system.

Functions:

- Manage user details
- Add and update health-related content
- Monitor system activities
- Ensure proper functioning of the system

Emergency Alert Module

This module ensures user safety during critical situations.

Functions:

- Send emergency alerts
- Share location details
- Contact emergency numbers

3.2.1 Health Monitoring and Reminder System

Health monitoring is a key component of the FemCare system. It is used to track and manage women's health conditions, especially menstrual cycles and related health activities.

The system allows users to enter health-related data and stores it in the database. Based on this data, the system provides reminders and notifications to help users maintain proper health tracking. This process helps users stay aware of their health status and manage their routine effectively.

The system uses simple data processing techniques to analyze user inputs and generate useful outputs such as reminders and health tips.

Health monitoring provides several benefits:

- Accurate tracking of health conditions
- Timely reminders for important dates
- Improved health awareness

Additionally, the reminder system ensures that users do not miss important health-related activities by sending notifications at the right time.

3.3 Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation of how data moves within a system. It shows how data is processed, stored, and transferred between different modules.

DFDs help in understanding how the FemCare system works and how different components interact with each other.

3.3.1 DFD Level 0

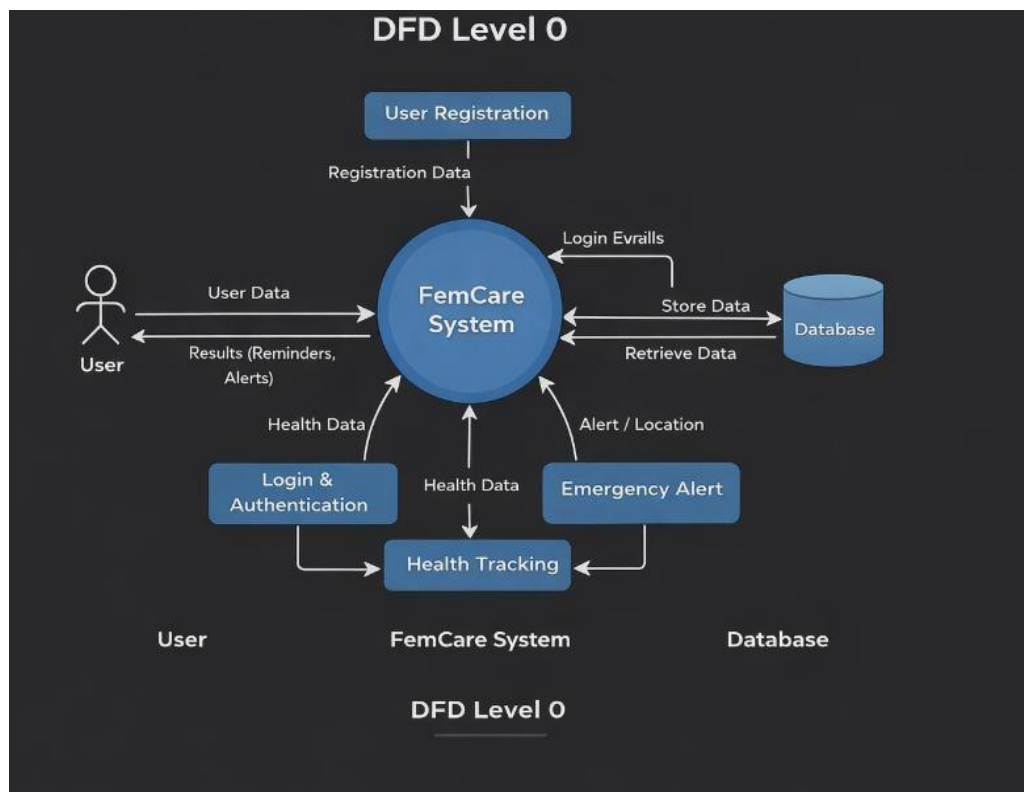
It shows the interaction between:

- **User**
- **FemCare System**
- **Database**

Main processes include:

- User Registration
- Login and Authentication
- Health Tracking
- Emergency Alert
- Data Storage

The system collects user data, processes it, and stores it securely in the database while providing necessary outputs like reminders and alerts.



3.3.1 DFD Level 1 –User Module

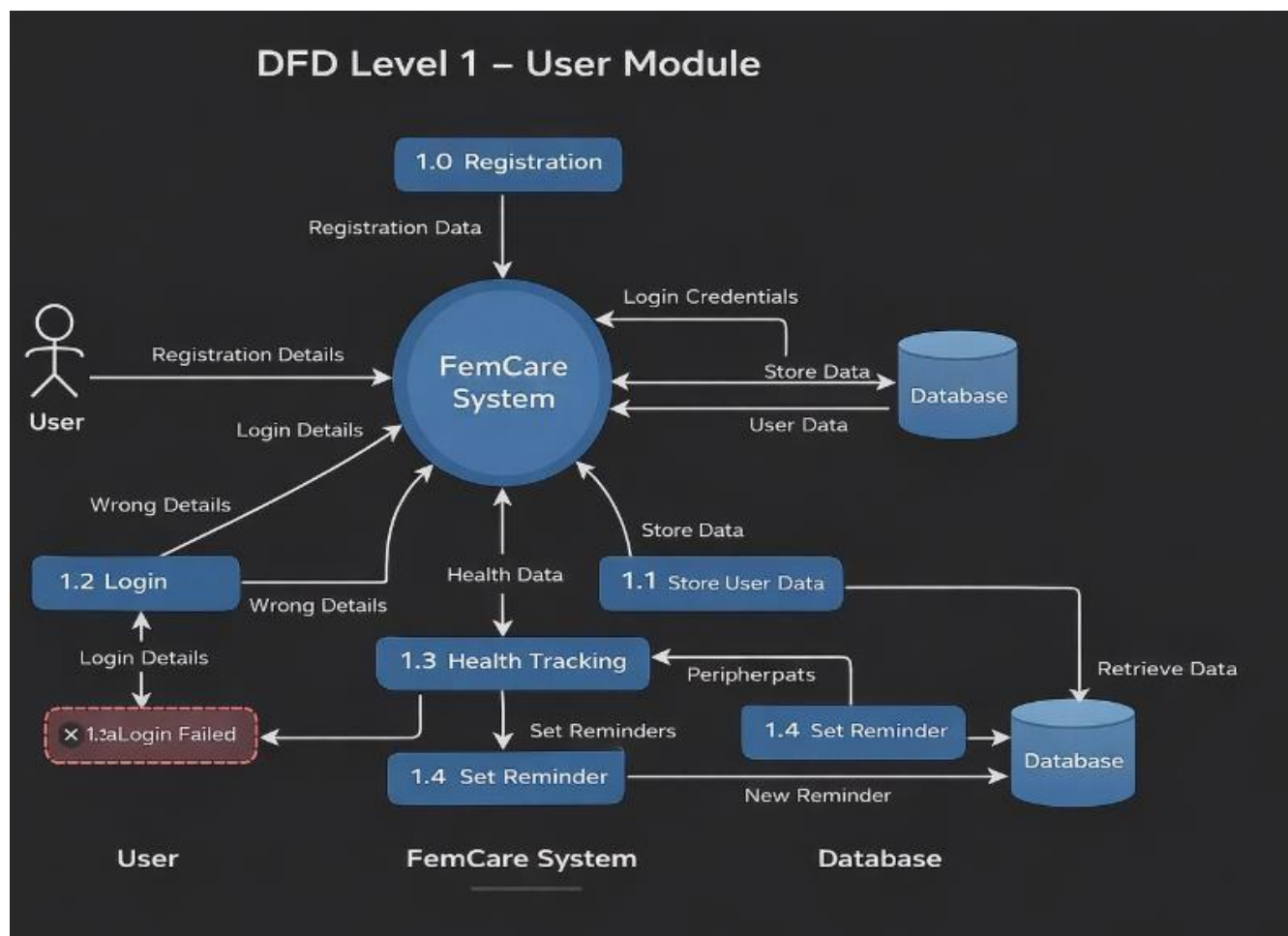
DFD Level 1 provides a detailed view of the main processes in the FemCare system, focusing on user interactions such as registration, login, health tracking, and emergency features. This level explains how data flows between the user, system, and database during different operations.

□ **User Registration**

- The user enters personal details such as name, email, and phone number
- The system stores the data in the database

□ **User Login**

- The user enters login credentials
- The system verifies the details with stored data

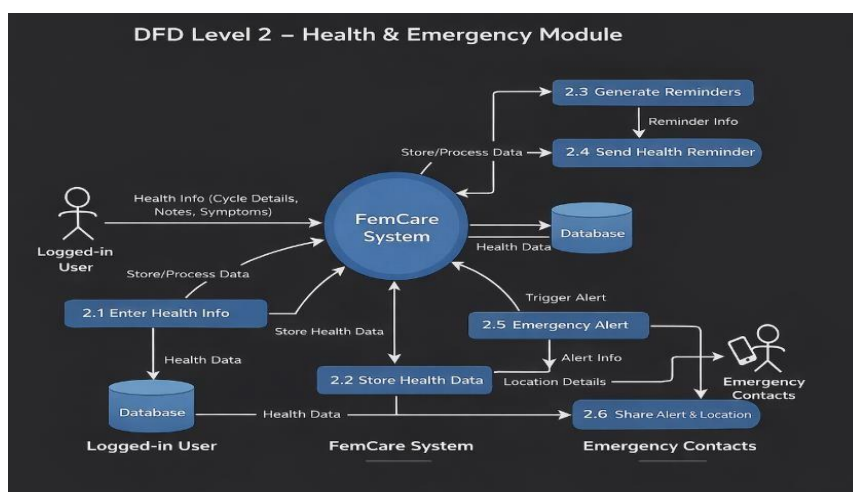


3.3.2 DFD Level 2 –Health and Emergency

DFD Level 2 provides a more detailed explanation of the core functionalities of the FemCare system, focusing on health tracking and emergency alert processes.

This level shows how the system processes user data, generates reminders, and handles emergency situations effectively.

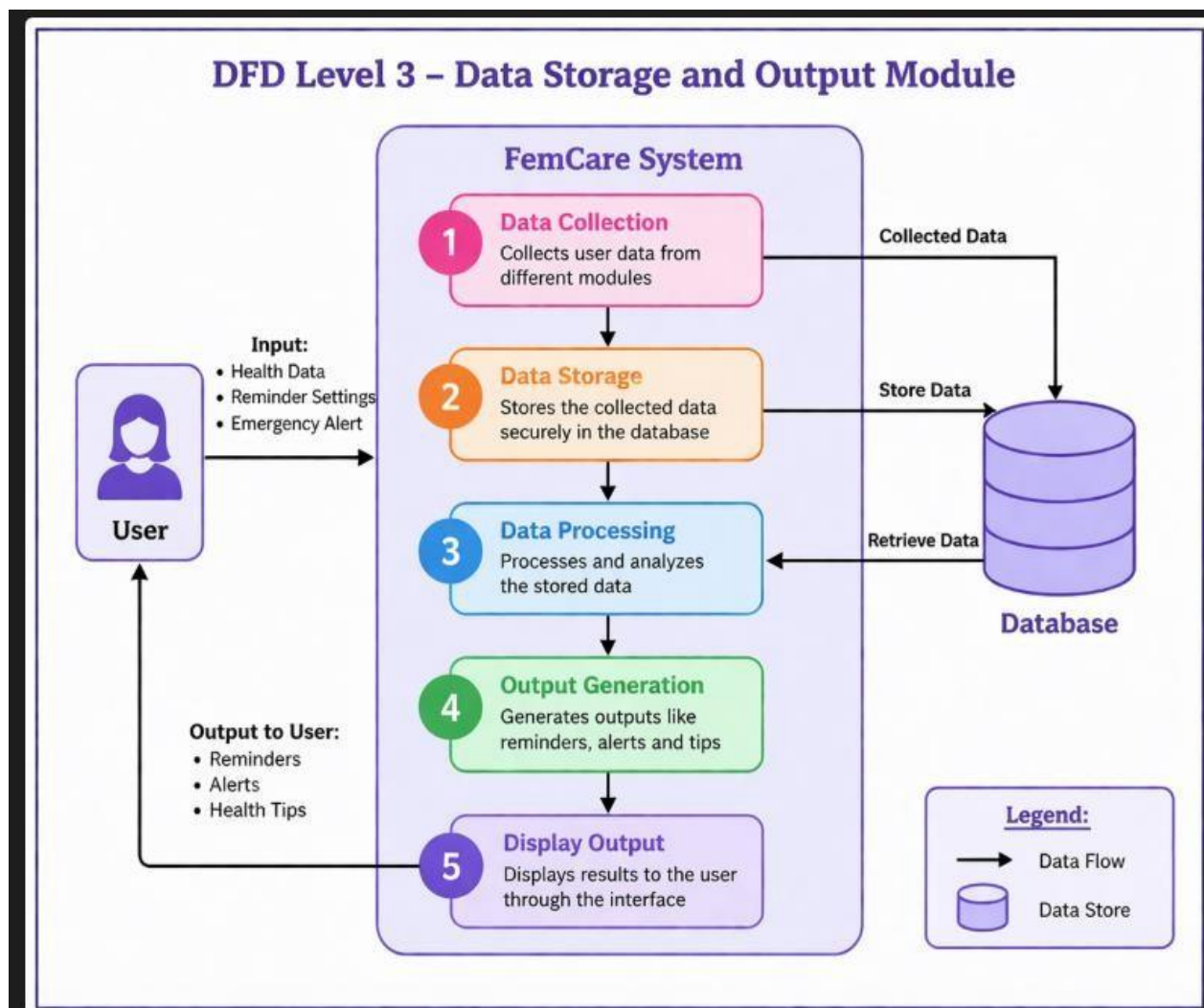
This process ensures strong Health & Emergency



3.3.3 DFD Level 3 – Data Storage & Output

- 4 DFD Level 3 provides a detailed view of how data is stored, processed, and displayed in the FemCare system. It focuses on how user data is managed and how the system generates outputs such as reminders, alerts, and health information.

This module ensures the **one-person-one-vote principle**.



4.1 Advantages

The FemCare system provides several advantages that improve women's health management and safety.

- Provides easy health tracking for menstrual cycles and related conditions
- Sends timely reminders for important health activities
- Ensures quick emergency support through alert system
- Improves health awareness by providing useful tips and information

4.2 Requirement Specification

4.1.1 Hardware Requirements

- Computer or laptop with webcam
- Minimum 4 GB RAM
- Processor: Intel i3 or higher
- Internet connectivity

4.1.2 Software Requirements

- Operating System: Windows
- Programming Language: Python
- Frontend: HTML, CSS, JavaScript
- Backend Framework: Flask
- Database: MySQL
- Machine Learning and Natural Language Processing

CHAPTER 4

IMPLEMENTATION AND RESULTS

This chapter describes the implementation of the **FemCare: Women Health and Safety Support System** and the results obtained during testing. The system was developed to provide health tracking, reminders, and emergency support for users.

The implementation includes different stages such as user registration, login, health data entry, reminder generation, and emergency alert functionality. Each module was tested to ensure proper working and accuracy.

4.1 Results of Face Detection

Face detection is the first step in the biometric authentication process. The system uses computer vision techniques to detect the presence of a human face in the image captured by the webcam.

During the voter authentication stage, the system activates the camera and scans the image frame to identify the facial region. Once the face is detected, the system extracts the facial features required for further recognition.

The face detection process was implemented using **OpenCV-based algorithms**, which accurately detect faces in real-time. The system successfully detects the voter's face under normal lighting conditions and different facial orientations.

4.2 Results

The registration and login module allows users to securely access the system.

Observed Results:

- Users were able to register successfully
- Login process worked correctly with valid credentials
- Invalid login details were rejected
- Data was stored securely in the database

This ensures secure access to the system.

4.1 Results of Health Tracking

The health tracking module helps users monitor their health conditions.

Observed Results:

- Users were able to enter health data easily
- Data was stored and retrieved correctly
- System provided accurate tracking of menstrual cycles
- Reminders were generated based on user input

This improves health awareness and management

4.2 Results of Reminder System

The reminder system notifies users about important health activities.

Observed Results:

- Reminders were sent on time
- Notifications helped users maintain health schedules
- System worked efficiently without delays

4.3 Results of Emergency Alert System

The emergency module provides quick support during critical situations.

Observed Results:

- Emergency alerts were sent successfully
 - Location sharing worked properly
 - Users were able to contact emergency numbers quickly
- This ensures user safety and quick response.

4.4 System Performance

The overall system performance was tested in a web environment.

Observed Results:

- System is fast and responsive
- User interface is simple and easy to use
- All modules worked smoothly
- No major errors were found

CHAPTER 5

DISCUSSION AND CONCLUSION

This chapter summarizes the overall results of the **FemCare: Women Health and Safety Support System**, discusses its limitations, and suggests future improvements. The system was developed to provide a combined solution for women's health tracking and safety support.

5.1 Key Findings

The implementation of the FemCare system shows that combining health tracking and emergency features in one platform is effective and useful.

Findings:

- Users can easily track their health data
- Reminder system helps in maintaining health routines
- Emergency alert feature provides quick support
- The system is simple and user-friendly
- Data is stored and managed efficiently

These results show that the system improves both health awareness and safety.

5.2 GitHub Link of the Project

The complete source code of the project, including the frontend, backend, and modules, has been uploaded to GitHub for reference and future development.

GitHub Repository Link: <https://github.com/sirisha1697/AICW-Femcare-Project>

5.3 Video Recording of Project Demonstration

A demonstration video of the project has been recorded to show the working of the system, including voter registration, OTP verification, face recognition authentication, and the voting process.

Project Demonstration Video Link: https://drive.google.com/file/d/1AiZN-R7S-aGfQ9S-HaPYB4cW5wYz9vT7/view?usp=drive_link

5.4 Limitations

Although the FemCare system works effectively, it has some limitations:

- Requires a **stable internet connection** to access all features
- Currently available only as a **web-based application** (no mobile app)
- Accuracy depends on **user input data**
- Limited advanced features like **AI-based suggestions**
- Emergency alert may depend on **network availability**
- Basic security features (can be improved further)

5.5 Future Work

Although the FemCare system works effectively, it has some limitations:

- Requires a **stable internet connection** to access all features
- Currently available only as a **web-based application** (no mobile app)
- Accuracy depends on **user input data**
- Limited advanced features like **AI-based suggestions**
- Emergency alert may depend on **network availability**
- Basic security features (can be improved further)

5.6 Conclusion

The **FemCare: Women Health and Safety Support System** was developed to address the challenges related to women's health management and personal safety. The system provides an integrated platform where users can track their health, receive timely reminders, and access emergency support when needed.

The implementation of the system shows that combining health tracking and safety features in a single application improves convenience, awareness, and user experience. The system is simple, efficient, and easy to use, making it suitable for everyday use.

Overall, FemCare is a reliable solution that helps women manage their health and ensures safety. With further enhancements and advanced features, the system can be improved and used on a larger scale in the future.

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